

ROS 2 Engineering Curriculum
Volume 1: Fundamentals

Self-Study Edition

May 29, 2026

Contents

Chapter 1

Learning Roadmap

This volume takes a learner from zero knowledge of ROS 2 to the point where they can create packages, nodes, publishers, subscribers, services, actions, and launch files.

Chapter 2

What is ROS 2?

2.1 Why Robotics Needs Middleware

Robots consist of many independent subsystems: sensors, perception, planning, control, and actuators. ROS 2 provides a common communication framework.

2.2 ROS 1 vs ROS 2

Topics:

- DDS
- Discovery
- Security
- Real-time support
- Embedded support

Exercises

Explain why a robot should be split into multiple nodes.

Chapter 3

Installing ROS 2

Ubuntu installation workflow, environment setup, workspace creation, and verification.

Chapter 4

ROS 2 Architecture

Layers:

1. Application
2. rclcpp/rclpy/rclc
3. rcl
4. rmw
5. DDS
6. Network Transport

Detailed discussion of responsibilities of each layer.

Chapter 5

Nodes

Node lifecycle:

1. Process creation
2. ROS initialization
3. Node creation
4. Communication setup
5. Execution
6. Shutdown

Lab: Create a node in C++ and Python.

Chapter 6

Topics

Publisher/subscriber communication model.

Lab: Create a temperature publisher and subscriber.

Quiz: Difference between synchronous and asynchronous communication.

Chapter 7

Messages

Message definitions, standard interfaces, custom messages, and serialization basics.

Lab: Create a custom message package.

Chapter 8

Services

Request-response communication.

Lab: Build a calculator service.

Chapter 9

Actions

Goal, feedback, result model.

Lab: Build a countdown action server.

Chapter 10

Parameters

Runtime configuration management.

Lab: Create configurable nodes.

Chapter 11

Launch Files

Python launch system, node orchestration, parameter injection, remapping.

Lab: Launch three communicating nodes.

Chapter 12

Development with C++

roscpp fundamentals, node classes, publishers, subscribers, services, actions, timers, parameters.
Programming exercises included.

Chapter 13

Development with Python

rcipy fundamentals, callback design, timers, services, actions.
Programming exercises included.

Chapter 14

Workspace and Build System

colcon, ament, package.xml, CMakeLists.txt, dependency management.

Lab: Build a multi-package workspace.

Chapter 15

Debugging and Introspection

Commands:

```
ros2 node list  
ros2 topic list  
ros2 topic echo  
ros2 topic info  
ros2 doctor
```

Lab: Investigate a running ROS graph.

Chapter 16

Mini Projects

Project 1: Robot health monitor.

Project 2: Sensor dashboard.

Project 3: Multi-node telemetry system.

Chapter 17

Final Assessment

Written questions:

1. Explain the ROS graph.
2. Explain publishers and subscribers.
3. Compare services and actions.
4. Describe message serialization.
5. Describe package dependencies.

Practical exam: Create a complete ROS 2 application from scratch.